Causeway

Technical & Economic Whitepaper

AI-native causal and execution infrastructure
for prediction markets.

Market Structure | Causal Intelligence | Vault Layer | Governance

A disciplined infrastructure layer between event evidence, market structure, and user-governed execution.

Version 1.0 | May 2026 | Causeway Visual Identity aligned

0. Important Notice

This whitepaper describes a software architecture and economic mechanism direction for Causeway. It is not investment, legal, tax, or trading advice. It does not solicit any transaction and does not promise liquidity, yield, prediction accuracy, execution quality, or profit.

Any future vault, basket, ETF-like, or mispricing-assessment capability must be interpreted as infrastructure design language. Legal classification, operating permissions, investor suitability, and jurisdiction-specific compliance require separate review.

0. Contents

No.	Chapter
1	Executive Summary
2	Market Structure Thesis
3	Prediction Market Infrastructure Gap
4	Causeway System Architecture
5	Outcome-Token Native Data Model
6	Source Object and Real-Time Information Layer
7	Market Matching Engine
8	Causal Intelligence Loop
9	Dislocation Assessment Framework
10	Human-Governed Execution Layer
11	Self-Hosted AI Control Plane
12	Open Vault Layer
13	Vault Risk and NAV Framework
14	Governance, Audit, and Compliance Boundaries
15	Technical Roadmap and Milestones
16	Limitations and Non-Goals
A	Data Model Summary
B	API Lifecycle Summary
C	Dislocation Variables
D	Vault Parameter Template
E	Audit Event Examples
F	Risk Matrix

1. Executive Summary

Strategic Interpretation

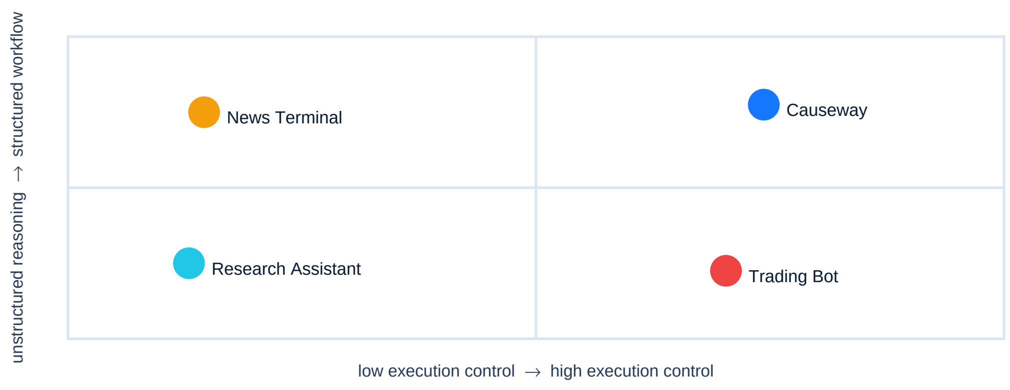
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Causeway's strategic position is not to use AI to help users predict markets, but to build the information-processing, causal-reasoning, execution-preparation, and portfolio-expression infrastructure for prediction markets. Prediction markets have already transformed real-world events into tradable probabilities, yet a verifiable, auditable, and extensible middle layer is still missing between event evidence and related markets, between market prices and outcome tokens, and between reasoning results and user-confirmed orders.

Causeway's long-term value is to integrate these broken links into a continuous workflow: information is standardized into source objects, markets are organized as outcome-token graphs, AI reasoning is constrained to verifiable candidate sets, execution is decomposed into preview, capability checks, signature, and confirmation, and portfolio exposure is further structured through open vaults. It is neither a prediction oracle nor a trading bot, but a system layer that can move prediction markets from page-level trading products toward institutional-grade intelligence infrastructure.

Frame	Design Position
Problem	Prediction-market interfaces expose prices, but not the structured workflow required to move from evidence to related markets, outcome tokens, risk checks, and monitored execution.
Mechanism	Causeway combines local market synchronization, outcome-token native data, causal scripts, order preparation, source matching, and vault-ready portfolio primitives.
Controls	AI output is bounded by candidates, schemas, validators, preview TTLs, user confirmation, and audit trails.
Output	An infrastructure path from analyst workflow to intelligence layer, self-hosted AI, and open vaults.

Positioning Map



2. Market Structure Thesis

Strategic Interpretation

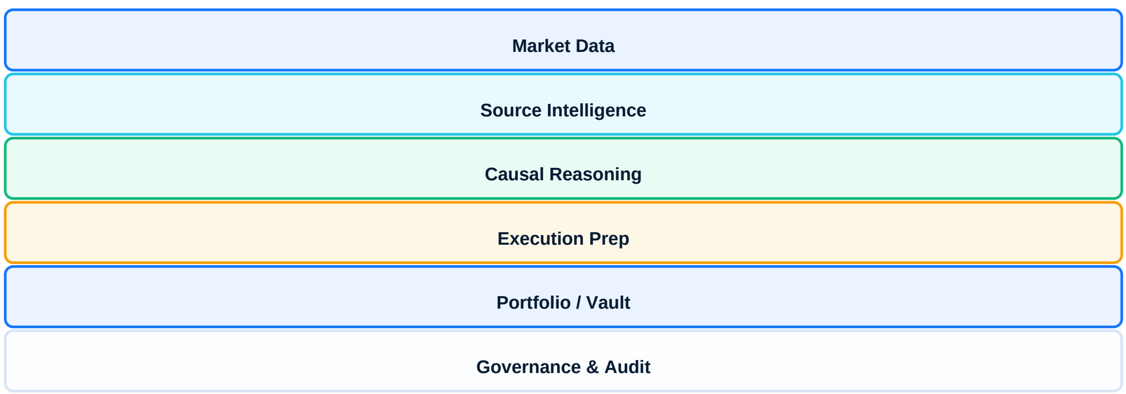
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

The core asset of a prediction market is not a single market page, but the constantly changing relationship network among events, narratives, prices, liquidity, and outcome tokens. Major real-world events rarely affect only one market; they propagate through policy, macro, sports, industry, on-chain, and social narratives into multiple related markets. Most current interfaces expose point prices but cannot express this cross-market transmission structure.

Causeway's strategic entry point is to make implicit event-transmission relationships explicit. Users should not only see how much a market is priced at, but also which markets an event thesis may affect, which outcome token is exposed, how strong the relationship is, whether the information is already priced in, and whether executable liquidity exists. This graph perspective is the common foundation for market matching, causal reasoning, dislocation scoring, and vault-based portfolio construction.

Frame	Design Position
Problem	A single real-world event can affect multiple markets, categories, and outcome tokens, but the relation is rarely explicit.
Mechanism	Causeway models event theses as roots that branch into related markets and tradable outcomes.
Controls	The graph is constrained by real market identifiers, tradability, liquidity, and audit history.
Output	Users can inspect how one thesis propagates through a market network before preparing orders.

Layered System Architecture



3. Prediction Market Infrastructure Gap

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Prediction markets already provide trading entry points, but they still lack the infrastructure layer that connects information, judgment, execution, and governance. Today, completing a full workflow requires users to discover related markets, read external information, assess price lag, verify outcome tokens, refresh order books, estimate slippage, control position size, and preserve decision evidence. These steps rely heavily on personal experience and are difficult to scale, review, or institutionalize.

Causeway fills the layer of a prediction-market operating system: it standardizes information ingestion, market matching, causal reasoning, execution preparation, risk controls, and audit records. Once this gap is systematized, prediction-market participants can move from manual browsing and ad hoc judgment to sustainable, reusable, and governable workflows. This is the key distinction between Causeway and market pages, news terminals, or ordinary AI assistants.

Frame	Design Position
Problem	Information latency, fragmentation, outcome-token complexity, liquidity constraints, and unmanaged AI outputs create operational risk.
Mechanism	Causeway introduces source objects, market matching, causal scripts, capability gates, and audit-first execution.
Controls	No real order is submitted without preview, capability checks, wallet authorization, and explicit user confirmation.
Output	A professional workflow that transforms market discovery into reviewable, executable, and monitorable state.

4. Causeway System Architecture

Strategic Interpretation

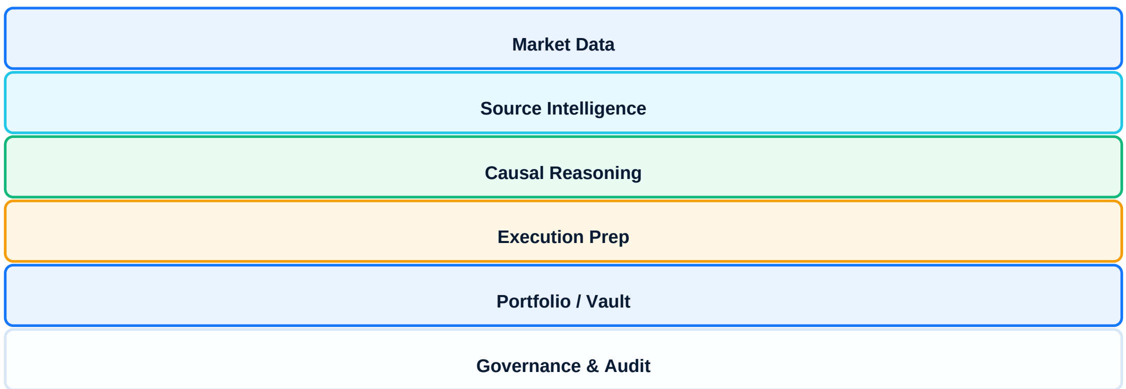
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Causeway's architecture must adhere to layered design because errors in prediction-market systems often arise from layer confusion: treating information as evidence, reasoning as trading advice, price as executable result, or AI output as user authorization. A layered architecture gives each component a clear responsibility and boundary: the data layer maintains factual state, the intelligence layer generates candidates and reasoning, the execution layer handles preview and confirmation, the portfolio layer expresses long-term exposure, and the governance layer records and constrains system behavior.

The strategic meaning of this architecture is extensibility. When Causeway later connects real-time information sources, self-hosted AI, dislocation scoring, or open vaults, it does not need to rebuild the system; it can add capabilities on top of existing layers. Each layer can be evaluated, replaced, audited, and governed independently, preventing the product from becoming an unexplained AI black-box trading system.

Frame	Design Position
Market Data	Events, markets, outcomes, prices, order books, source objects, snapshots, and raw payloads.
Intelligence	Candidate retrieval, source matching, causal reasoning, and dislocation assessment.
Execution	Order input, preview, signature preparation, submit, reconciliation, and audit.
Portfolio / Vault	Positions, open orders, vault mandates, basket construction, risk budgets, and NAV-style reporting.

Layered System Architecture



5. Outcome-Token Native Data Model

Strategic Interpretation

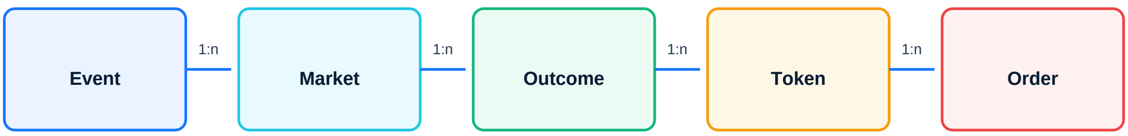
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Outcome tokens are the smallest credible unit of every Causeway capability. Prediction markets appear to be natural-language questions on the surface, but real execution occurs at the tokenId level. If the system only understands market titles or Yes/No labels, it will introduce semantic errors in multi-outcome markets, sports lines, range markets, candidate markets, and special-rule markets, ultimately affecting reasoning, orders, and auditability.

Causeway must therefore strictly separate Event, Market, Outcome, and Outcome Token, and consistently reference stable identifiers across AI reasoning, user edits, order previews, and vault construction. This data model is not an engineering detail; it is a strategic moat. Whoever can correctly understand outcome tokens has the right foundation to build higher-level intelligence, portfolio, and execution infrastructure on prediction markets.

Frame	Design Position
Problem	Real markets are not fixed Yes/No instruments; outcome labels are display text, not durable identifiers.
Mechanism	Every tradable object is normalized to marketId, outcomeId, and CLOB tokenId.
Controls	AI references must resolve to existing candidate markets and outcome tokens.
Output	A stable bridge between market data, AI recommendations, user edits, order previews, and audit records.

Outcome-Token Native Data Model



6. Source Object and Real-Time Information Layer

Strategic Interpretation

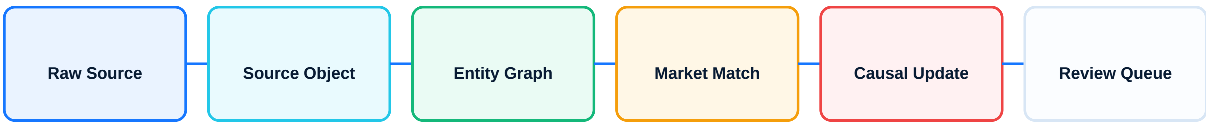
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Real-time information sources are the fuel for Causeway's future intelligence layer, but raw information cannot directly enter models and trading workflows. News, social media, on-chain events, sports data, macro releases, and official announcements all differ in origin, timeliness, noise, and credibility. Without standardization, the system cannot explain why a market was matched, why a score increased, or why a script requires renewed review.

The strategic role of the Source Object is to convert the external world into a computable, traceable, and governable evidence layer. Each item should record source, timestamp, entities, tags, summary, confidence, affected markets, evidence type, and raw payload. In this way, Causeway does not merely see information; it turns information into part of market matching, causal updates, dislocation assessment, and audit records.

Frame	Design Position
Problem	Raw news, social posts, on-chain events, sports feeds, and official announcements are noisy and hard to audit.
Mechanism	Each source object stores origin, timestamp, entities, tags, summary, confidence, freshness, evidence type, affected markets, and raw payload.
Controls	Source provenance, reliability, retention, and confidence decay are explicit.
Output	Real-time information becomes queryable, replayable, and matchable to markets.

Source Ingestion Pipeline



7. Market Matching Engine

Strategic Interpretation

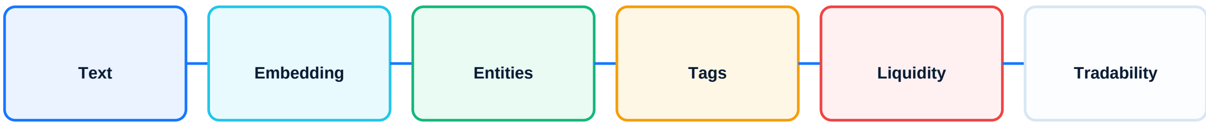
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

The market matching engine is the decisive step through which Causeway moves from an information product into market-intelligence infrastructure. A simple information feed tells users what happened; the matching engine answers which real tradable markets and outcome tokens the event corresponds to. This is more complex than ordinary search because it must simultaneously consider semantic relevance, entity relationships, event hierarchy, historical resonance, market activity, liquidity, and tradability.

A professional matching system should not merely output a result list; it should provide matching rationale and confidence. Causeway's long-term competitiveness will come from explainable matching: users can see why a source affects a market, why it points to a specific outcome, and why the result is downgraded because of insufficient liquidity. Only this type of match can become reliable input for causal reasoning, dislocation scoring, and execution preparation.

Frame	Design Position
Problem	Open-ended AI may invent markets or miss execution constraints.
Mechanism	Matching combines text retrieval, embeddings, entity extraction, tag overlap, event hierarchy, historical co-movement, liquidity filters, and tradability filters.
Controls	Each match returns reason, confidence, liquidity status, tradability status, and referenced token identifiers.
Output	A ranked candidate set for causal reasoning and review.

Matching Score Decomposition



8. Causal Intelligence Loop

Strategic Interpretation

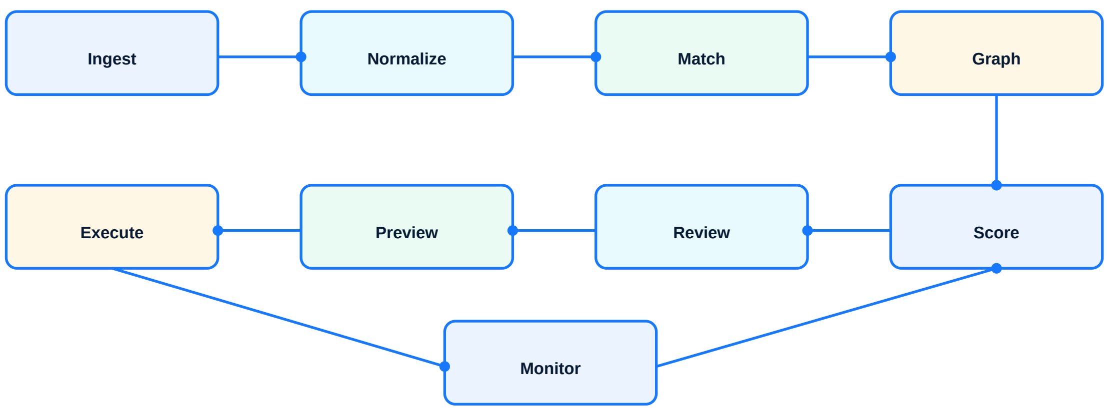
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Causeway's intelligence should not take the form of one-off prompts, but of a continuously operating system loop. One-time AI outputs are poorly suited to real-time information changes, market price updates, order-state feedback, and user behavior data. Real market intelligence must keep receiving new information, updating market relationships, reassessing dislocations, triggering review, observing execution outcomes, and feeding those results back into system evaluation.

The strategic significance of the causal intelligence loop is that Causeway becomes a state-maintenance system rather than a suggestion generator. It continuously maintains relationships among source objects, market graphs, causal scripts, order states, and portfolio risks. As the system runs, audit records, user edits, failure cases, fills, and market changes become material for evaluating AI, matching, and risk-control mechanisms.

Frame	Design Position
Problem	Single AI calls are hard to govern, refresh, evaluate, or audit.
Mechanism	Source ingestion, normalization, market matching, causal graph update, dislocation scoring, review, preview, execution, monitoring, and feedback operate as a loop.
Controls	Human checkpoints remain before order preview and before real submission.
Output	A continuously evaluable system for market reasoning.

Causeway Intelligence Loop



9. Dislocation Assessment Framework

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Market dislocation assessment must remain professionally restrained. It should not claim to discover guaranteed opportunities; it should help users identify which markets deserve priority review. A divergence between event evidence and market price is only practically meaningful when market relevance is strong, price response is delayed, liquidity is acceptable, execution cost is controlled, and the information has not expired.

Causeway's dislocation score should therefore be decomposed into explainable variables: evidence strength, market relevance, price lag, liquidity-adjusted edge, execution risk, and confidence decay. The value of this framework is not to decide for the user, but to show why something deserves attention. It makes the review queue more rational while preventing the system from sliding into black-box signal calling or autonomous trading narratives.

Frame	Design Position
Problem	Evidence may be relevant but untradable, illiquid, stale, noisy, or already priced in.
Mechanism	A composite score separates evidence strength, market relevance, price lag, liquidity-adjusted edge, execution risk, and confidence decay.
Controls	Score bands produce Ignore, Monitor, Review, and High-priority Review states rather than autonomous trade signals.
Output	A review queue with explainable components.

Dislocation Score Decomposition

$$D = w1 \cdot E + w2 \cdot R + w3 \cdot L + w4 \cdot Q - w5 \cdot X - w6 \cdot T$$

Ignore	Monitor	Review	High Review
0.00-0.30	0.30-0.55	0.55-0.75	0.75-1.00

10. Human-Governed Execution Layer

Strategic Interpretation

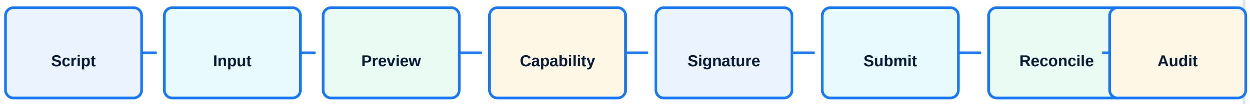
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

The execution layer is Causeway's most important safety boundary. AI can help discover related markets, generate causal scripts, and propose candidate actions, but it cannot replace user confirmation or bypass order preview, wallet signatures, and risk checks. Prices and order books in prediction markets change quickly; without preview TTLs, order-book refresh, capability checks, and idempotent submission, automation will amplify risk.

Causeway's execution strategy is to enhance user control rather than eliminate it. dry_run lets users rehearse strategies without submitting real orders; preview shows current cost and failure reasons; capability gates state whether real execution is available; audit logs preserve evidence for every step. This execution layer can support more complex workflows in the future while preserving non-custodial and human-confirmed product boundaries.

Frame	Design Position
Problem	Trading workflows fail when stale prices, missing order books, duplicate submits, or opaque capability states are ignored.
Mechanism	Script selection, order input, order preview, capability check, signature preparation, user confirmation, submit, reconciliation, and audit.
Controls	Preview TTL, idempotency keys, partial-failure handling, dry_run mode, and real capability gates.
Output	Safe rehearsal, clear failure states, and auditable real execution.

Order Lifecycle Sequence



11. Self-Hosted AI Control Plane

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

As Causeway enters real-time information matching, user strategy storage, portfolio monitoring, and vault management, self-hosted AI will become a core control plane rather than an optional capability. The reason is not merely cost or performance, but privacy, latency, model governance, and auditability. Which markets a user monitors, how they modify scripts, what positions they hold, and which sources they subscribe to may all become highly sensitive strategy data.

The self-hosted AI control plane should include an inference gateway, model registry, prompt registry, schema registry, output validator, evaluation framework, audit log, and fallback router. Only then can Causeway answer critical questions: which model generated which result under which input, did the result pass validation, did an upgrade change the output distribution, and can errors be replayed and repaired. This is the difference between a generic AI application and professional infrastructure.

Frame	Design Position
Problem	User strategies, watchlists, source matching preferences, and position context become sensitive as the system matures.
Mechanism	Inference gateway, model registry, prompt registry, schema registry, source-matching model, market-reasoning model, output validator, evaluation harness, audit logger, fallback router.
Controls	Versioned outputs, regression tests, benchmark suites, drift checks, and replayable audit logs.
Output	Lower dependency risk and stronger control over market-reasoning quality.

Self-Hosted AI Control Plane

12. Open Vault Layer

Strategic Interpretation

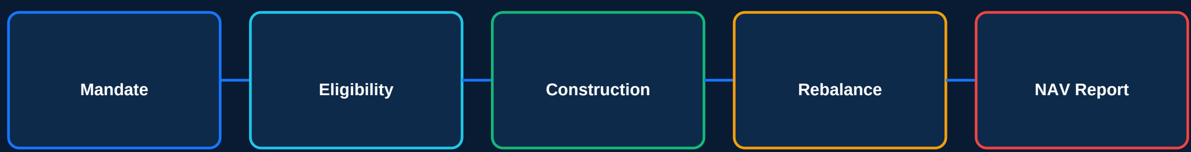
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Open vaults are the key module through which Causeway moves from a user workflow tool toward prediction-market portfolio infrastructure. Individual markets are suitable for active judgment, but many users and institutions need thematic, rules-based, and transparent basket exposure, such as positions around elections, macro policy, sports seasons, industry regulation, or on-chain events.

The professionalism of the vault layer comes from mechanism, not packaging. Each vault should define mandate, eligible markets, position-construction rules, rebalancing policy, risk budget, NAV-style reporting, redemption boundaries, and governance parameters. Prediction-market ETF-ization means providing an experience similar to basket exposure and transparent reporting; it does not mean legal equivalence to an ETF, and it does not imply guaranteed return.

Frame	Design Position
Problem	Individual market selection does not scale into thematic exposure without governance and risk accounting.
Mechanism	Vault mandate, eligibility rules, position construction, rebalancing policy, risk budget, NAV-style reporting, redemption boundary, and governance.
Controls	Eligibility filters, exposure caps, liquidity thresholds, drawdown limits, and disclosure.
Output	ETF-like basket experience without implying legal ETF classification or guaranteed return.

Open Vault Lifecycle



ETF-like means basket exposure and reporting experience, not legal classification or guaranteed return.

13. Vault Risk and NAV Framework

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

The central difficulty of prediction-market vaults is valuation and risk presentation. Traditional asset portfolios already have mature valuation frameworks, while prediction-market outcomes may face low liquidity, thin order books, market closures, unresolved events, stale prices, and rule disputes. Simply aggregating several outcomes into a basket would obscure the real risk.

Causeway's vault framework must introduce risk budgets and NAV-style reporting from day one. NAV calculation should consider current prices, open orders, unsettled positions, resolved markets, fees, liquidity haircuts, and stale-data adjustments. Risk budgets should include position caps, market caps, category caps, correlation limits, liquidity caps, and drawdown limits. Only then is ETF-ization a professional portfolio expression rather than a marketing slogan.

Frame	Design Position
Problem	Prediction-market baskets can hide concentration, liquidity, stale data, unresolved positions, and settlement risk.
Mechanism	VaultRiskBudget = PositionCap + MarketCap + CategoryCap + LiquidityCap + DrawdownLimit + CorrelationLimit.
Controls	NAV incorporates current outcome price, open orders, unsettled positions, resolved markets, liquidity haircuts, fees, and stale-data adjustments.
Output	Portfolio-style reporting that remains honest about market-specific risk.

14. Governance, Audit, and Compliance Boundaries

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Causeway's institutional potential depends on governance capability. Model versions, prompts, schemas, source classifiers, matching weights, dislocation scoring weights, vault mandates, and execution permissions all affect what users see, what they review, and what they execute. If these parameters are not traceable, the system cannot be trusted or reviewed when disputes arise.

Causeway must therefore treat governance and audit as core product capabilities rather than background logs. Every inference, match, user edit, order preview, submission, failure, and vault rebalance should be recorded. At the same time, the whitepaper must define compliance boundaries clearly: Causeway does not provide investment advice, does not guarantee returns, does not trade autonomously, does not custody private keys, and uses ETF-like only as a product-experience and reporting analogy, not as a legal classification.

Frame	Design Position
Problem	AI, matching weights, scoring weights, vault mandates, and execution permissions all create governance risk.
Mechanism	Versioned prompts, models, schemas, source classifiers, score weights, vault parameters, and audit retention.
Controls	No investment advice claim, no guaranteed returns, no autonomous AI trading, no private key custody, ETF-like not legal ETF classification.
Output	A system whose decisions can be reconstructed and challenged.

15. Technical Roadmap and Milestones

Strategic Interpretation

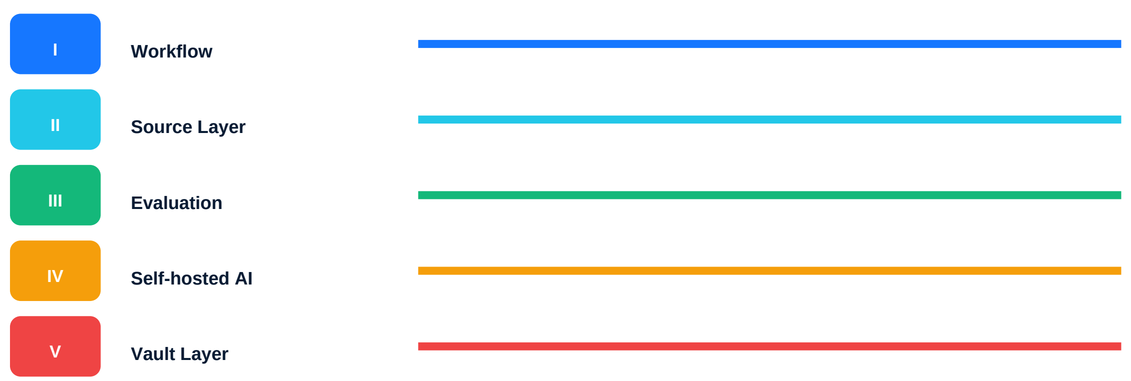
Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Causeway's roadmap must advance according to capability dependencies rather than feature stacking. The first phase should harden the outcome-token data model, market synchronization, causal scripts, and order preview; if the underlying data is unstable, any AI reasoning or vault product will rest on fragile foundations. The second phase should introduce real-time information sources and Source Objects, the third should build the matching engine and dislocation scoring, the fourth should advance self-hosted AI and evaluation systems, and the fifth should expand open vaults and ecosystem interfaces.

The professionalism of this roadmap lies in the fact that each stage produces verifiable capability: whether data synchronizes correctly, whether matches are explainable, whether scores can be backtested, whether models are auditable, and whether vaults can report transparently. It avoids presenting Causeway as a one-shot, all-in-one product and instead frames it as prediction-market infrastructure built layer by layer.

Frame	Design Position
Phase I	Workflow layer: market sync, outcome model, causal scripts, order preview.
Phase II	Real-time source layer: source objects, ingestion, provenance, freshness.
Phase III	Matching and dislocation layer: matching engine, score bands, review queues.
Phase IV	Self-hosted AI and vault layer: private inference, model governance, vault mandates, NAV reporting.
Phase V	Protocol and ecosystem layer: partner APIs, external vault integrations, governed automation boundaries.

Roadmap Swimlane



16. Limitations and Non-Goals

Strategic Interpretation

Chapter guide: strategic positioning, infrastructure logic, and product boundary.

Limitations and non-goals strengthen rather than weaken the whitepaper's credibility. Prediction markets inherently involve uncertainty, liquidity constraints, event-interpretation disputes, and user risk tolerance. Any system that promises guaranteed prediction accuracy, guaranteed returns, or fully autonomous trading is not suitable to be defined as long-term infrastructure. Causeway must draw these boundaries proactively.

Causeway should not be understood as a prediction oracle, yield product, investment adviser, or automated trading bot. Its objective is to improve the quality of information structuring, market discovery, causal reasoning, execution preparation, risk disclosure, and audit governance. Users retain judgment, authorization, and risk responsibility. Precisely because it does not promise to replace user decision-making, Causeway is more likely to become professional, sustainable, and extensible prediction-market infrastructure.

Frame	Design Position
Non-goals	No guaranteed prediction accuracy, no guaranteed profit, no guaranteed fill quality, no autonomous AI trading, no private-key custody.
Data boundaries	V1 should not rely on unmanaged external sources inside AI inference.
Legal boundary	ETF-like describes experience and reporting, not legal classification.
Product boundary	Dislocation scoring is a review-priority tool, not an automatic buy/sell signal.

A. Data Model Summary

Object	Key Fields	Purpose
SourceObject	sourceId, sourceType, origin, timestamp, entities, confidence, rawPayload	Evidence provenance and matching input
PolymarketOutcome	marketId, outcomeId, clobTokenId, price, bestBid, bestAsk	Atomic tradable object
CausalScript	rootOutcomeId, graphJson, selections, audit state	Editable reasoning workspace
OrderIntent	executionMode, previewJson, riskJson, capability	User-specific order package
Vault	mandate, eligibility, riskBudget, navPolicy	Rule-based basket container

B. API Lifecycle Summary

Stage	Representative APIs	Control
Auth	POST /auth/nonce, POST /auth/verify	Wallet signature and session
Markets	GET /markets, GET /markets/:id, GET /orderbook	Local data and fresh book
Inference	POST /inference-runs, GET /inference-runs/:id	Async status and validation
Scripts	GET /scripts/:id, PATCH selections	Human editability
Orders	POST /orders/preview, prepare-signature, submit	Preview TTL, capability, idempotency

C. Dislocation Variables

Variable	Meaning	Control
E	Evidence strength	Source reliability and evidence type
R	Market relevance	Entity, tag, semantic, and historical match
L	Price lag	Observed delay between evidence and price response
Q	Liquidity-adjusted edge	Potential mismatch after spread and depth
X	Execution risk	Slippage, book depth, failure probability
T	Confidence decay	Freshness and noise degradation

D. Vault Parameter Template

Parameter	Example	Boundary
Mandate	Election-related policy basket	Defines eligible theme
Market Eligibility	active, acceptingOrders, min liquidity	Prevents untradeable exposure
Position Cap	max 5% per outcome token	Limits concentration
Rebalance	weekly or event-triggered	Defines adjustment cadence
Disclosure	no guaranteed yield, no legal ETF claim	User-facing boundary

E. Audit Event Examples

Event	Required Fields	Reason
InferenceRunCreated	userId, rootOutcomeId, model, promptVersion	Reconstruct AI context
SourceMatched	sourceId, marketId, outcomeId, confidence	Explain matching
OrderPreviewed	intentId, bookTime, expiresAt, riskJson	Validate execution basis
OrderSubmitted	intentId, idempotencyKey, result	Prevent duplicate ambiguity
VaultRebalanced	vaultId, oldWeights, newWeights, ruleVersion	Govern portfolio changes

F. Risk Matrix

Risk	Severity	Mitigation
Stale data	High	Freshness thresholds and preview TTL
Model hallucination	High	Candidate constraints and output validation
Liquidity failure	High	Depth-aware preview and capability gates
Regulatory ambiguity	High	Clear non-advice and product classification review
User overreliance	Medium	Disclosure, review states, no autonomous submission